

AI Validation Log

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Project: HimShakti Food Processing Raw Material Expiry & Inventory Waste Dashboard

Section 1 – AI Prompt

I am working on a Business Intelligence project for HimShakti's Food Processing Raw Material Expiry & Inventory Waste Dashboard. The dataset contains the following columns:

Batch_ID, Raw_Material, Entry_Date, Expiry_Date, Quantity_KG, Remaining_Quantity_KG, Production_Date, Finished_Good, and Shelf_Life_Days.

Based on this dataset, suggest five important business questions that can help improve inventory management, reduce raw material wastage, and support better business decisions.

Section 2 – AI Generated Insight Questions

Q-1 Which raw materials are approaching their expiry dates and require immediate attention?

Q-2 Which months received the highest number of raw material batches?

Q-3 Which raw materials have the highest remaining inventory quantity?

Q-4 Which finished goods have the highest average shelf life?

Q-5 Is there any relationship between total inventory quantity and remaining inventory quantity?

Section 3 – AI Validation Table

AI Insight Question	Evaluation	Reason
Which raw materials are approaching their expiry dates and require immediate attention?	Accurate	The dataset contains Expiry_Date, so expiring batches can be identified easily.
Which months received the highest number of raw material batches?	Accurate	Entry_Date is available, making monthly trend analysis possible.
Which raw materials have the highest remaining inventory quantity?	Accurate	Remaining_Quantity_KG is available and supports inventory analysis.
Which finished goods have the highest average shelf life?	Accurate	Shelf_Life_Days and Finished_Good columns allow this calculation.
Is there any relationship between total inventory quantity and remaining inventory quantity?	Partially Accurate	The relationship can be analyzed, but it does not directly explain inventory efficiency without additional business context.

Section 4 – AI Claims Edited (Before vs After)

This section highlights a few AI-generated statements that were reviewed and corrected after validating them against the actual dashboard dataset. The purpose was to ensure that all business insights were supported by data rather than assumptions.

Example 1

AI Claim (Before):

High remaining inventory quantity indicates overstocking of raw materials.

Verified Statement (After):

A high remaining inventory quantity only shows that more stock is currently available. It does not necessarily indicate overstocking because additional information such as production demand, consumption rate, and procurement planning is required to reach that conclusion.

Example 2

AI Claim (Before):

The month with the highest number of batch entries represents the highest customer demand.

Verified Statement (After):

The month with the highest batch entries indicates increased procurement or inventory activity. Customer demand cannot be confirmed without sales or order data.

Example 3**AI Claim (Before):**

Finished goods with longer shelf life are always the most profitable.

Verified Statement (After):

A longer shelf life improves storage flexibility and reduces the risk of expiry, but profitability depends on several additional factors such as production cost, selling price, and customer demand.